Steca TR A503 TTR

5 inputs, 3 outputs

Alongside the two Steca TR A501 T and Steca TR A502 TT controllers, Steca TR A503 TTR is the third product in this series, and is presented in a compact housing design which means it can be used universally.

Steca TR A503 TTR expands the functionality of the series in particular with the additional functions found in the service of detailed system control. The 'system pressure monitoring' controller function enables additional system control together with an external pressure sensor. An additional controller output is also available, which enables individual programmability for different functions.

As with the other products in this series, a specially designed switched-mode power supply is used to achieve the greatest possible efficiency and economical operation. This reduces self-consumption to a minimum. The variable input voltage range ensures universal and worldwide use.

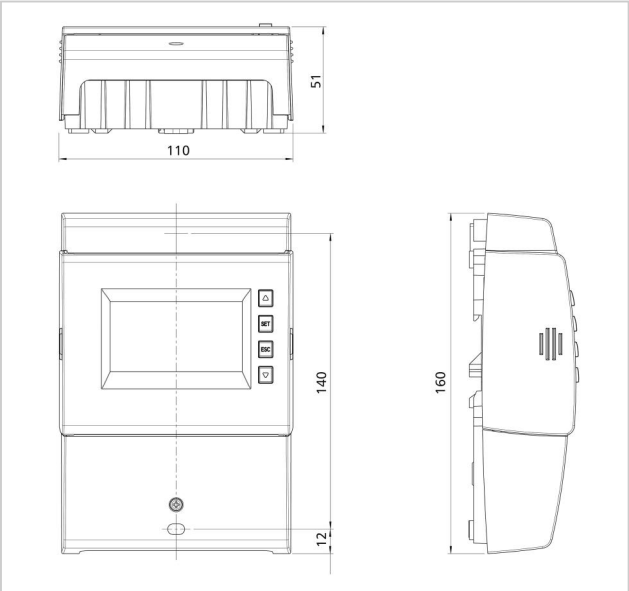
The Steca TR A503 TTR has two speed-controllable triac outputs, which can optionally also be used as PWM outputs for controlling high-efficiency pumps, and a relay switching output for additional controller functions. Five inputs for temperature or pulse recording round off the equipment features.

The integrated heat meter enables the solar yields to be recorded with imputed values. For higher demands, this can of course also be done with the help of an external pulse generator or by using an additional Grundfos Direct Sensor™.

Steca TR A503 TTR monitors and controls solar thermal systems with up to two differently oriented collector arrays or a maximum of two domestic hot water or buffer tanks. With the help of the third controller output, the additional controller functions can be used individually.

Eleven pre-programmed equipment systems enable versatile use. Important system monitoring and safety functions are also carried out by the Steca TR A503 TTR, as are special fault indicators for quickly rectifying faults.

As with the other types in this series, the Steca TR A503 TTR uses the newly developed electronic overload control and protects the device against overloading and installation errors.



	TR A503 TTR
System voltage	115 V AC … 230 V AC, 50 Hz / 60 Hz
Standby	0.76 W
Temperature control class	I
Energy efficiency	1 %
Number of inputs	5
Inputs	4 x temperature (Pt1000), 1 x temperature (Pt1000) or pulse
Additional inputs	1 x Grundfos Direct Sensors™ (temperature / flow rate), 1 x Grundfos Direct Sensors™ (temperature / pressure)
Number of outputs	3
Outputs	2 x triac for speed control (R1, R2), max. 250 W (230 V) or PWM control signal for pump speed (PWM R1, PWM R2); 1 x switch output relay (R3), 2 A, 115 V AC … 230 V AC
Additional output	1 x potential-free switching output for the safety extra-low voltage
Hydraulic schemes	11
Ambient temperature	0 °C … +50 °C
Degree of protection	IP 22 / DIN 40050 [without front panel: IP 20]
Dimensions (X x Y x Z)	110 x 160 x 51 mm [0.00 x 0.00 x 0.00 inch]
Weight	370 g [0.00 oz]

Product features
- Compact, multipart designer casing
- Installation versions: Solar pump stations, wall installation, mounting rails
- Wave packet (Triac) and pulse width modulation (PWM) ensure electronic RPM control
- High level of operational safety through fault diagnosis
- Hours-of-operation logger
- Software update possible
- Daily pump start
- Screw terminals allow universal and rapid installation
- Low power consumption thanks to universal and widerange switched-mode power supply
- Variable input voltage range for worldwide controller deployment
- Electronic overloading control and protection

Displays
- Multifunction graphical LC display with backlighting
- Animated representation of the systems and operating states

Operation
- Non-verbal menu navigation
- Manual switch for manual, auto, off

Functions
- Data logging to a MicroSD card
- Heat quantity (Grundfos Direct Sensors™, pulse generator, determination)
- Display showing CO_2 savings
- System pressure monitoring (e.g. Grundfos Direct Sensors™)
- Reduction of stagnation phases
- Active cooling (e.g. to avoid stagnation)
- Holiday (storage tank recooling)
- Circulation (controlled by temperature / time)
- Back-up heating
- Solid fuel boiler
- Storage tank quick charge
- Thermostat
- Differential thermostat
- Interval / tube collector
- Anti-freeze
- Display storage tank top
- Alarm output